Spencer M. Pollard

PhD Applied Mathematics
Master of Science Computer Science
Bachelors of Science Physics & Computer Science
University of California, Davis

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Spencer Pollard
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Education

- **University of California, Davis** — *2025-Present*
  PhD Applied Mathematics

- **California State University, Chico** — *2024-2025*
  Master of Science - Computer Science — GGPA: 3.967/4.0
  Thesis: A Comparison of Kalman Filtering and Deep Learning for State Estimation in Chaotic Systems
  Committee: Dr. Samuel B. Siewert, Dr. Richard C. Tillquist, Dr. Nicholas J. Nelson

- **California State University, Chico** — *2021-2024*
  Bachelor of Science - Physics (Professional Physics) — CGPA: 3.922/4.0
  Bachelor of Science - Computer Science
  Minor: Mathematics
  Summa Cum Laude

Awards

- **Helen Quinn Award** APS Far West 2024 — *December 2024*
- **Josie Otwell Student Assistant Award** — *August 2024*
- **Lieutenant Robert Merton Rawlins Merit Award** — *August 2024*
- **Physics Summer Research Institute (PSRI) Fellowship** — *June 2024*
- **Michael R. McGie Service Scholarship** — *August 2023*
- **Physics Summer Research Institute (PSRI) Fellowship** — *June 2023*
- **Chico STEM Connections Collaborative Fellowship** — *June 2023*
- **Chico State Dean's List Recipient** — *2021 - 2025*
- **Member of Sigma Pi Sigma Honor Society** — *April 2023*
- **Member of Alpha Lambda Delta Honor Society** — *April 2023*

Presentations

- "A Comparison of Kalman Filtering and Deep Learning for State Estimation in Chaotic Systems", S. Pollard, Chico State CSCI 300 Seminar

- "A Comparison of Physical Models with Kalman Filtering to Deep Learning Models for Complex System State Estimation and Prediction", S. Pollard, S. Siewert, SPIE Defense + Commercial Sensing Conference Signal Processing, Sensor/Information Fusion, and Target Recognition XXXIV

- "An Experience Report on Teaching Quantum Key Distribution to Incoming College Freshmen", S. Pollard, A. Attarwala, J. Raigoza, American Society for Engineering Education Southeastern Section (2025)

- "Computational Modeling of Femtosecond Coherence Spectroscopy for Vibronic-Exciton Model Dimers", S. Pollard, P. C. Arpin, American Physical Society Far West Section (2024)

- "Development of a Mobile Application to Enhance Scientific Reasoning Abilities in a GE Course for Non-STEM Majors", S. Pollard, J. Pechkis, H. Pechkis, B. Dixon, American Physical Society Far West Section (2023)

- "Development of a Mobile App for "Physics of Music" Course", S. Pollard, J. Pechkis, H. Pechkis, B. Dixon, CSC^2 Research Symposium (2023)

Publications

- Pollard, S. M., Attarwala, A., and Raigoza, J. "An Experience Report on Teaching Quantum Key Distribution to Incoming College Freshmen", ASEE Southeastern Section Conference, 2025.
- Pollard, S. M. and Siewert, S. "A Comparison of Physical Models with Kalman Filtering to Deep Learning Models for Complex System State Estimation and Prediction", SPIE Defense + Commercial Sensing Conference Signal Processing, Sensor/Information Fusion, and Target Recognition XXXIV

Experience

- **Graduate Research Assistant** — *June 2026 - Present*
  Los Alamos National Lab — Los Alamos, NM
- **Math Department Teaching Assistant** — *September 2025 - Present*
  University of California, Davis — Davis, CA
  - Lead discussion sections and grade quizzes and exams for undergraduate mathematics courses.
- **Physics Department Student Assistant** — *August 2024 - May 2025*
  California State University, Chico — Chico, CA
  - Rebuilt quantum optics experiments after the new Natural Sciences building was built for the physics Advanced Laboratory course
  - Performed spontaneous parametric down-conversion and the existence of photons by splitting blue laser light using a nonlinear BBO crystal and counting coincidences of the entangled red photon pairs
  - Performed single-photon interference using a Mach-Zehnder interferometer by measuring resultant spectra
- **Physics Research Assistant** — *June 2024 - August 2024*
  California State University, Chico — Chico, CA
  - Worked alongside Dr. Paul Arpin continuing the work based on his paper Theoretical model of femtosecond coherence spectroscopy of vibronic excitons in molecular aggregates
  - Using Python and MATLAB, created spectra of amplitude quantum-beat frequency as a function of quantum-beat frequency and detection frequency for vibronic-exciton model dimers
  - Create spectra of amplitude quantum-beat frequency as a function of quantum-beat frequency and inter-molecular coupling energy.
- **Computer Science Graduate Research Assistant** — *June 2024 - August 2024*
  California State University, Chico — Chico, CA
  - Worked alongside Dr. Sam Siewert & Dr. Jaime Raigoza with bringing the University of Waterloo's Quantum Key Distribution classroom experiment to Chico State.
  - Made lesson plans and administered learning environment for students to learn about quantum key distribution through a hands-on activity.
  - Purchased materials, 3D printed, and built kits for groups of students to perform BB84 Quantum Key Distribution protocol in a studio physics laboratory setting.
- **Lead Undergraduate Research Assistant** — *June 2023 - May 2024*
  California State University, Chico — Chico, CA
  - Led development of a mobile app for the "Physics of Music" course as a research assistant.
  - Responsible for all aspects: software development, platform choice, and UI/UX design
  - Work closely with faculty mentors to manage the team, acquire new equipment, and aid individual team members with current tasks.
  - Mentor new team members to optimize application development.
- **Instructional Student Assistant** — *August 2022 - May 2024*
  California State University, Chico — Chico, CA
  - Served as Instructional Student Assistant, providing valuable support to the Physics, Math, Earth and Environmental Sciences, and Computer Science departments.
  - Conducted grading duties for the following courses: second-semester non-calculus based electricity and magnetism physics, second-semester calculus, Environmental Sensing, and Advanced Algorithms and Complexities. Assignments were graded by ensuring accurate assessment and timely feedback for students.
  - Fostered a supportive learning environment for students, promoting academic growth and success.
- **High School Geometry Teacher** — *September 2023 - June 2024*
  North Hills Christian School — Vallejo, CA
  - Provide students with daily lectures, homework, and assessments to provide meaningful feedback for student development

Research Projects

•Mobile App for "Physics of Music" General Education Course *June 2023 - May 2024*

Mobile app for California State University, Chico's "Physics of Music" course

- Faculty Advisors: Dr. Joseph Pechkis, Dr. Hyewon Pechkis, & Dr. Bryan Dixon
- Funded by NSF ($399,927.00): Using Mobile Apps to Enhance Students' Learning and Scientific Reasoning Skills in General Education Physics Courses
- Tools & technologies used: Dart, Flutter
- This app combines audio features from preexisting apps in a more student-friendly and educative manner. This app supports single-tone generation, power spectrum distribution for signal analysis, and an audio filter to extract the underlying wave from noisy input signals.

•Development of New Data Acquisition Software *November 2023 - January 2024*

Development of New Data Acquisition Software for Lidar: Raman-shifted Eye-safe Aerosol Lidar System

- Faculty Advisor: Dr. Shane Mayor
- Funded by NSF: Collaborative Research: Theoretical and observational investigations of multi-point Monin-Obukhov similarity in the convective atmospheric surface layer
- Tools & Technologies Used: Python, Tkinter, PyQT5, NIMAX, NIDAQmx
- Current LiDAR system runs on a 20+ year old system running LabVIEW code that is administered remotely and we would like to move away from LabVIEW
- Responsible for writing interfacing software between National Instruments Data Acquisition peripherals in Python and writing a GUI application to display and monitor current data acquisition processes
- Interface Moog Animatics SmartMotors by building custom cables and reading manufacturer's documentation for motors to behave and respond to user interaction on front-end UI

Technical Academic Projects

•EFieldFlow - Capstone *January 2024 - May 2024*

Computer Science Capstone for California State University, Chico's Computer Science Program

- Tools & technologies used: C#, Unity
- This desktop application takes inspiration from my previous E&M project where the user will graphically create a 3-D electric charge configuration to calculate the electric field at a point in space.
- Display a qualitative electric vector field around the charge configuration to help introductory physics students build intuition for E&M physics.

•OrganizeOrbit - Software Engineering *June 2023 - July 2023*

Final project for California State University, Northridge's Software Engineering course

- Tools & technologies used: HTML, CSS, JavaScript, Google Firebase, NodeJS
- This web application is to be a simple, space-themed to-do list primarily for those with ADHD. The web application comprises of users creating a to-do list of their daily tasks which are preserved in Google Firebase for future reference.

•Project E&M - Electricity and Magnetism *January 2023 - May 2023*

Final project for California State University, Chico's intermediate E&M course

- Tools & technologies used: Java, Apache NetBeans
- This desktop application allows the user to graphically create a 2-D electric charge configuration and calculate the electric field at a point in space.

Personal Projects

•Discord Game Server Manager *2022*

Custom Discord bot to manage a video game server

- Tools & technologies used: Python, DiscordPY
- This Discord bot is hosted on an 8GB Raspberry Pi 4 whose purpose is to allow my friends to host and play a video game together without the need for a dedicated person to host the game. The bot stays online waiting for user input to start and stop the server as well as run administrative commands within the game.

•Custom 9 Panel DDRxPIU Arcade Dance Pad *January 2019 - May 2019*

Combination of two arcade dancing rhythm games' dance pads

- This is the combination of four panels from "Dance Dance Revolution" and five panels "Pump It Up" into a single nine-panel dance pad. The dance pad was constructed out of wood for the frame and acrylic for each button panel. The button presses were made from custom button panels connected to a USB encoder to convert the electrical impulses from the button presses to be registered as a HID-compliant peripheral.

Technical Skills and Interests

Languages: Java, Python, C/C++, SQL, LaTeX, Intel IA-32 Instruction Set, MATLAB, Mathematica, LabVIEW, BASH, OCaml
Developer Tools: Git/GitHub, Visual Studio Code, Apache NetBeans, Jira
Frameworks: Flutter
Cloud/Databases: MySQL, Hadoop, Google Firebase
Areas of Interest: Embedded Systems, Scientific Computation, Robotics, Mathematical Modeling, Fluid Dynamics, Condensed Matter Physics, Computational PDEs

Professional Service to Community

•**Computer Science Tutor,** Department of Computer Science *August 2024 - May 2025*

•**Physics Tutor,** Society of Physics Students *August 2023 - May 2025*

•**Activities Coordinator,** Society of Physics Students *August 2023 - May 2024*